

1. Methods: Architecture choices

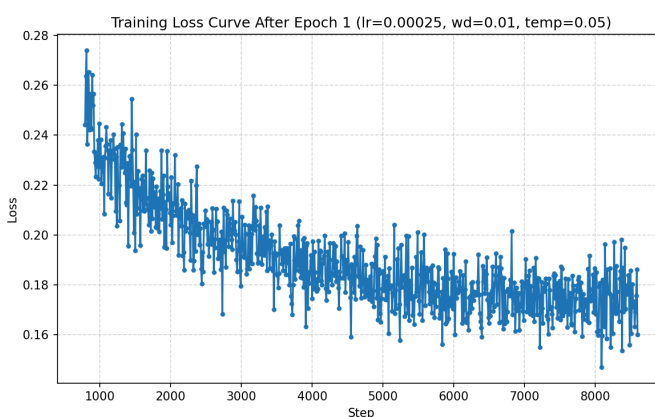
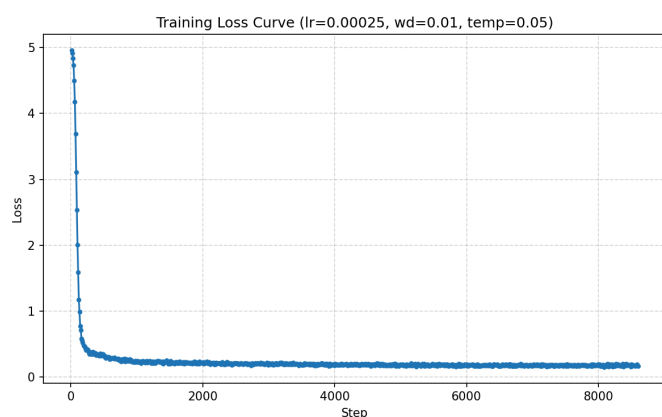
- **Vision encoder:** ResNet50 (`IMAGE_ENCODER="resnet"` in `config.py`) for stable training and manageable memory usage on 16 GB GPU.
- **Text encoder:** RoBERTa-base (`model/text_encoder.py`) with masked mean pooling over token embeddings.
- **Shared embedding space:** image/text embeddings are projected to `EMBED_DIM=512`.
- **Projection heads:** MLP heads (`Linear -> ReLU -> Linear`) on both towers.
- **Similarity + objective:** L2-normalized embeddings with learnable temperature (`log_temperature`), trained using symmetric contrastive loss (image-to-text CE + text-to-image CE, averaged).

2. Fine-tuning

2.1. Hyperparameters, run configs

- **Fine-tuning strategy:** freeze most pretrained weights; unfreeze only top layers (`UNFREEZE_IMAGE_BLOCKS=1`, `UNFREEZE_TEXT_LAYERS=1`).
- **Optimizer setup:** LR $2.5e-4$, weight decay 0.01 , temperature 0.05 .
- **Schedule:** cosine decay with warmup ratio 0.05 .
- **Training length:** 11 epochs.
- **Batching:** train batch 128 with grad accumulation 2 (effective 256), eval batch 256.
- **Validation snapshot (epoch 11):** eval loss 0.3245 ; top-1/top-5/top-10 $0.7873/0.9997/0.9999$; best checkpoint `checkpoint-6256`.

2.2. Training loss curves



- Training is stable: sharp early loss drop, then gradual improvement with moderate stochastic noise.
- No divergence is observed across the 11-epoch run.

3. Prompting Ablation experiments

3.1. Overview + Motivation

- ImageNet near 100% for fine-tuned model, but CIFAR performance is far behind
- This suggests issues with prompting strategy rather than fine-tuning
- So I shifted GPU resources to running prompting ablation experiments instead of hyperparameter-tuning experiments.
- Experiments ladder:

- Exp A: baseline templates
- Exp B: expanded prompt pack
- Exp C: class-aware prompt variants
- Ablated dimensions: prompt template set and class-specific aliases.
- Selection metric: weighted top-1 score in `eval.py` (default: CIFAR-10 **0.4**, CIFAR-100 **0.4**, ImageNet **0.2**; auto re-normalized when ImageNet is skipped).

3.2. ImageNet performance

- ImageNet was skipped in ablation scoring to reduce runtime and prioritize CIFAR-focused iteration.
- This trade-off is reasonable because final no-extra-ablation ImageNet performance is already near saturation (top-1 about **0.9986**).
- **Final fine-tuned model:** top-1 **0.9986**, top-5 **0.9997**, top-10 **0.9999**.
- (Baseline) **Pretrained model:** top-1 **0.0013**, top-5 **0.0078**, top-10 **0.0178**.

3.3. CIFAR performance across ablation sweeps (same table as before)

- Final choice: prompt-pack recipe without extra TTA/weighting (B/C tie; simpler config preferred).
- This is the prompt setup used for the final predictions.json generation

Setup	CIFAR-10 Top-1	CIFAR-100 Top-1	CIFAR-10 Top-2 / Top-3	CIFAR-100 Top-5 / Top-10
Pretrained weights, no prompt-pack	0.1462	0.0103	0.2770 / 0.3971	0.0561 / 0.1091
Pretrained weights, with prompt-pack	0.1370	0.0078	0.2449 / 0.3511	0.0455 / 0.0957
Final weights, Exp A (no prompt-pack)	0.4290	0.1161	0.6110 / 0.7180	0.2778 / 0.3905
Final weights, Exp B (with prompt-pack)	0.4379	0.1208	0.6226 / 0.7304	0.2853 / 0.3988
Final weights, Exp C	0.4379	0.1208	0.6226 / 0.7304	0.2853 / 0.3988

4. Lingering weaknesses analysis

4.1 Recurring failure patterns (post-eval artifacts)

Analysis of misclassified images shows consistent confusion pairs across ablation setups:

- **CIFAR-10:**
 - `automobile -> truck`
 - `airplane -> ship`
 - `cat <-> dog`

- horse -> deer, horse -> dog
- bird -> dog, bird -> cat
- frog -> deer, frog -> dog
- **CIFAR-100:**
 - keyboard -> clock
 - orange -> rose
 - man -> chimpanzee
 - worm -> snake
 - shark -> sea
 - streetcar -> house

The same over-predicted wrong labels also recur: **dog/deer/truck/ship** on CIFAR-10 and **sweet_pepper/porcupine/chimpanzee/otter/house** on CIFAR-100.

4.2 Why these errors persist

- Most failure cases are tiny (32x32) and visually ambiguous after resize/normalization.
- Errors are concentrated in semantically similar groups (vehicle-vehicle, mammal-mammal, object-object), indicating limited fine-grained discrimination under low resolution.
- Prompting improvements help overall metrics but do not materially change dominant confusion structure, especially on CIFAR-100.

4.3 Illustrative misclassification examples

CIFAR-10 examples: True: horse, Pred: dog / True: automobile, Pred: truck



CIFAR-100 examples: True: cloud, Pred: lamp / True: keyboard, Pred: clock

